

# One family. Many kinds of power.

An illustrated atlas of every announced M-series system on a chip, M1 through M6.

## 20 chips / 6 generations / 2020–2026

### Read two axes, not one

Generation changes the core design and capabilities. Tier changes the scale: CPU and GPU resources, memory capacity, bandwidth and packaging. A newer base chip need not replace an older Ultra for a large workload.

### How to read the plates

Every small square represents a core. Block positions, areas, cache layout and routing are illustrative. Core totals and documented die topology come from Apple. The diagrams are functional top-down schematics, not die photographs or measured floorplans. The memory strip is a shared pool, not on-die DRAM.

### What the numbers mean

This guide records maximum published chip configurations and historical advertised memory ceilings. It is not a list of every binned device configuration or a current buying guide. Performance ratios are not inferred from core counts. Empty tiers are not predicted products.

### Contents

2: Complete catalogue · 3–7: All 20 chip plates · 8–13: Detailed topology · 14–16: Architecture notes · 17: Building blocks · 18–20: Generations · 21–23: Experiments · 24–26: Sources

[Open the interactive atlas →](#)

# The complete catalogue

Maximum published configuration. Memory capacity is the historical advertised ceiling.

| Chip     | Date    | CPU | GPU | GB  | GB/s | NE | Ref. |
|----------|---------|-----|-----|-----|------|----|------|
| M1       | 2020-11 | 8   | 8   | 16  | —    | 16 | 1    |
| M1 Pro   | 2021-10 | 10  | 16  | 32  | 200  | 16 | 2    |
| M1 Max   | 2021-10 | 10  | 32  | 64  | 400  | 16 | 2    |
| M1 Ultra | 2022-03 | 20  | 64  | 128 | 800  | 32 | 3    |
| M2       | 2022-06 | 8   | 10  | 24  | 100  | 16 | 4    |
| M2 Pro   | 2023-01 | 12  | 19  | 32  | 200  | 16 | 5    |
| M2 Max   | 2023-01 | 12  | 38  | 96  | 400  | 16 | 5    |
| M2 Ultra | 2023-06 | 24  | 76  | 192 | 800  | 32 | 6    |
| M3       | 2023-10 | 8   | 10  | 24  | 100  | 16 | 7    |
| M3 Pro   | 2023-10 | 12  | 18  | 36  | 150  | 16 | 7    |
| M3 Max   | 2023-10 | 16  | 40  | 128 | 400  | 16 | 7    |
| M3 Ultra | 2025-03 | 32  | 80  | 512 | 819  | 32 | 8    |
| M4       | 2024-05 | 10  | 10  | 32  | 120  | 16 | 10   |
| M4 Pro   | 2024-10 | 14  | 20  | 64  | 273  | 16 | 10   |
| M4 Max   | 2024-10 | 16  | 40  | 128 | 546  | 16 | 10   |
| M5       | 2025-10 | 10  | 10  | 32  | 153  | 16 | 11   |
| M5 Pro   | 2026-03 | 18  | 20  | 64  | 307  | 16 | 12   |
| M5 Max   | 2026-03 | 18  | 40  | 128 | 614  | 16 | 12   |
| M5 Ultra | 2026-08 | 36  | 80  | 512 | 1200 | 32 | 13   |
| M6       | 2026-08 | 12  | 12  | 32  | 170  | 32 | 13   |

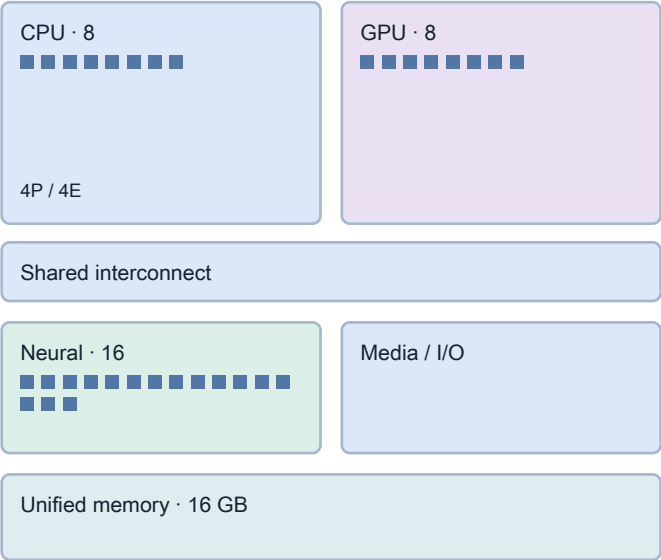
NE = Neural Engine cores. CPU and GPU counts are not performance units. M1 exact bandwidth is not stated in the cited launch source. M3/M4 Max lower bins can have 300/410 GB/s instead of 400/546 GB/s. M3 Ultra was advertised at up to 512 GB at launch; later sales configurations differ. M6 has two 16-core Neural Engines.

# Architecture plates / 1

S: super · P: performance · E: efficiency. Every square = one core. GPU neural units (M5/M6) are not counted as additional GPU cores.

## M1

2020-11 / maximum configuration

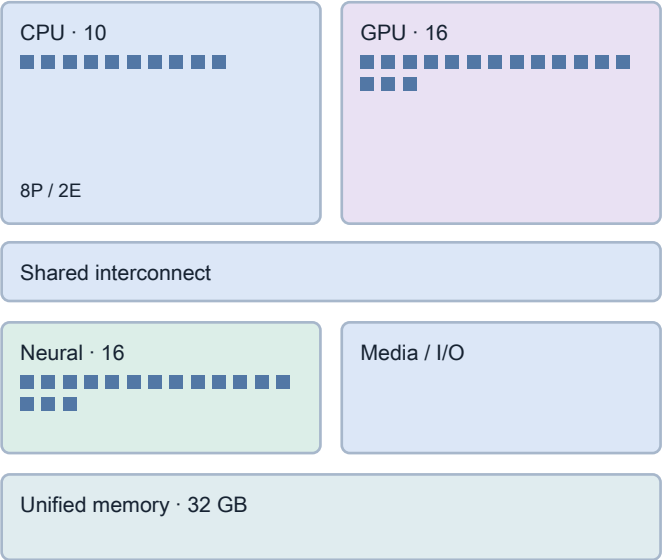


Bandwidth unspecified in source · 1 die(s)

Functional schematic; not to physical scale.

## M1 Pro

2021-10 / maximum configuration

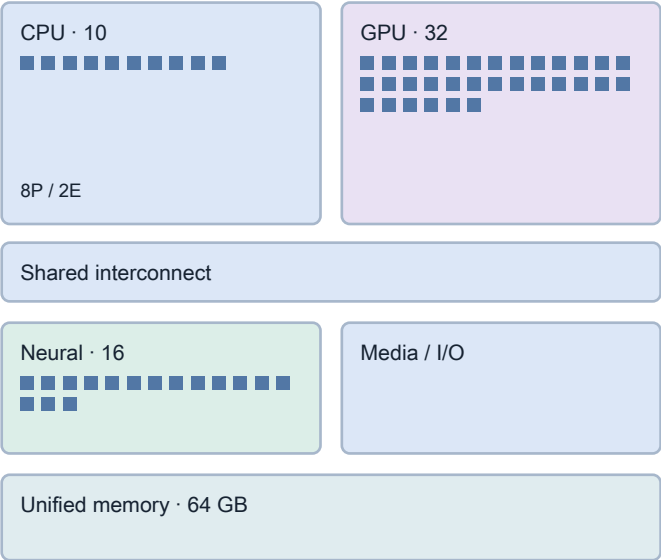


200 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M1 Max

2021-10 / maximum configuration

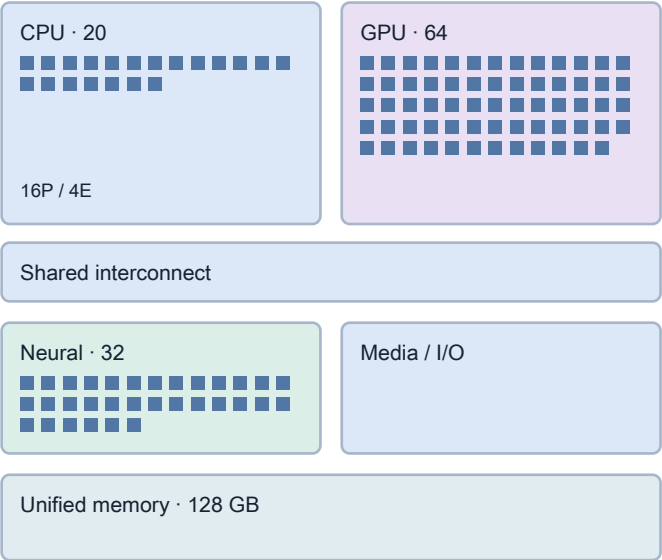


400 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M1 Ultra

2022-03 / maximum configuration



800 GB/s · 2 die(s)

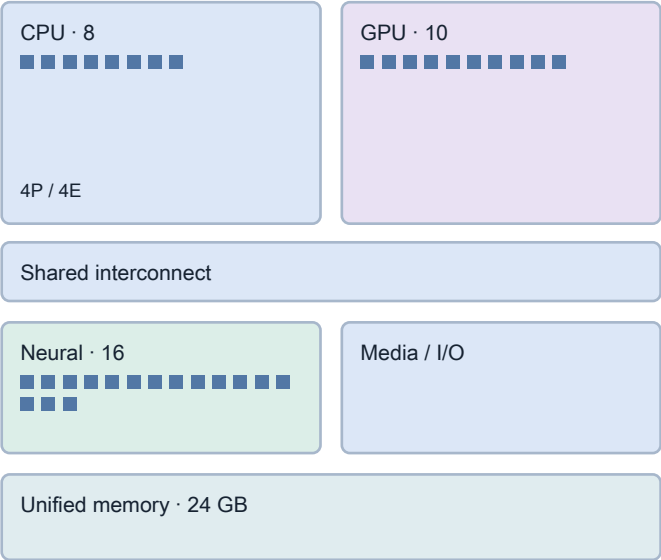
Functional schematic; not to physical scale.

# Architecture plates / 2

S: super · P: performance · E: efficiency. Every square = one core. GPU neural units (M5/M6) are not counted as additional GPU cores.

## M2

2022-06 / maximum configuration

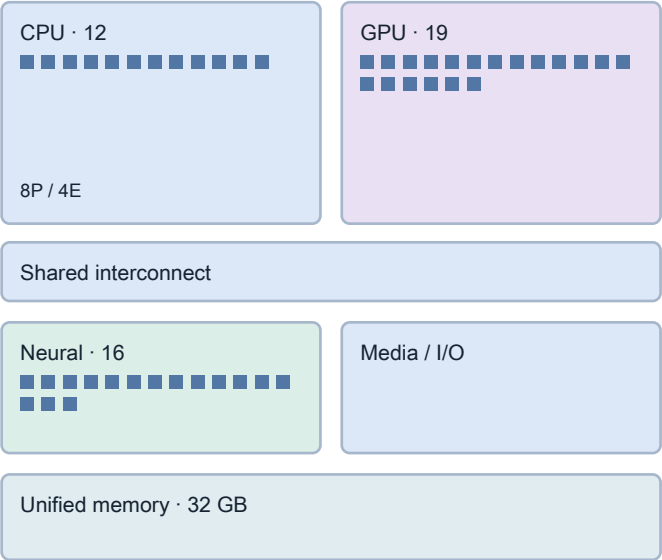


100 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M2 Pro

2023-01 / maximum configuration

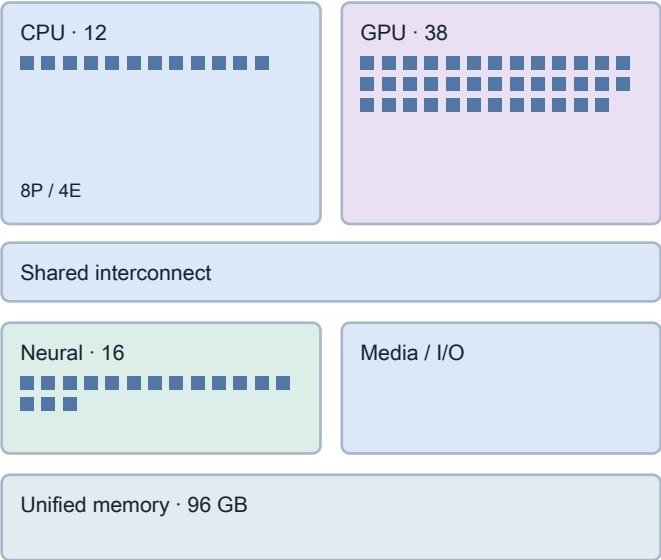


200 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M2 Max

2023-01 / maximum configuration

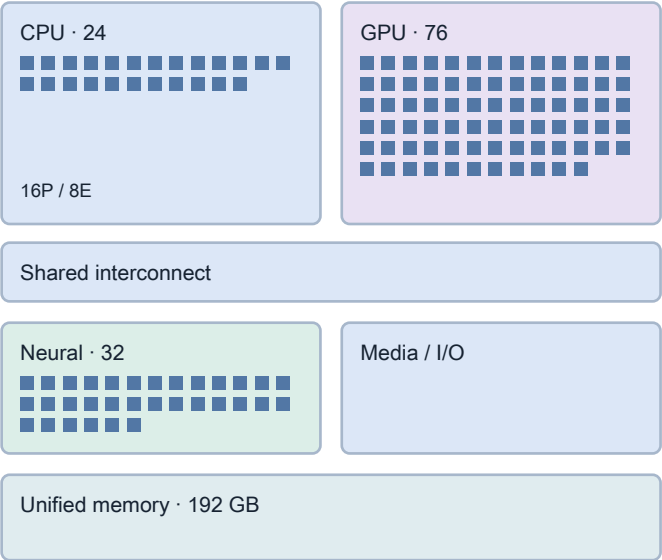


400 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M2 Ultra

2023-06 / maximum configuration



800 GB/s · 2 die(s)

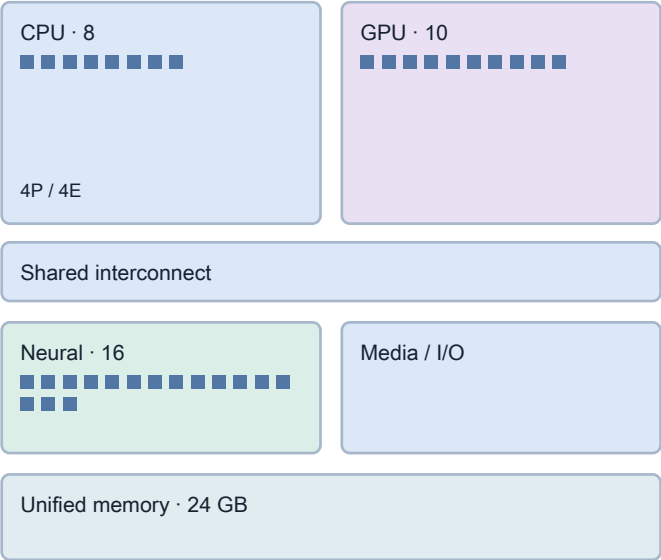
Functional schematic; not to physical scale.

# Architecture plates / 3

S: super · P: performance · E: efficiency. Every square = one core. GPU neural units (M5/M6) are not counted as additional GPU cores.

## M3

2023-10 / maximum configuration

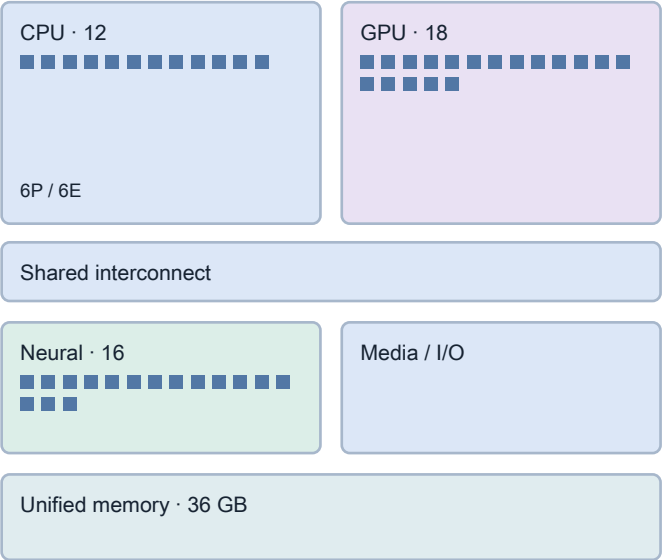


100 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M3 Pro

2023-10 / maximum configuration

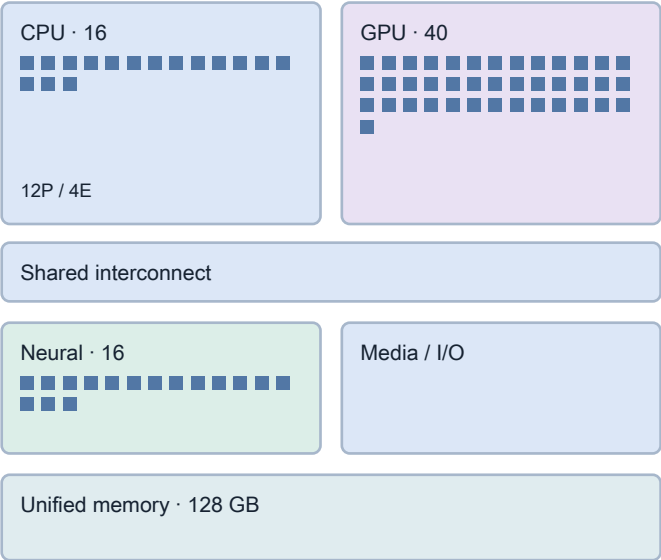


150 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M3 Max

2023-10 / maximum configuration

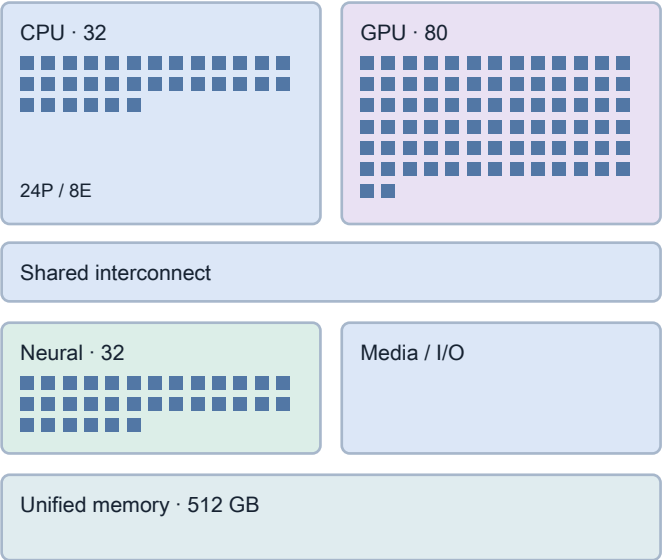


400 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M3 Ultra

2025-03 / maximum configuration



819 GB/s · 2 die(s)

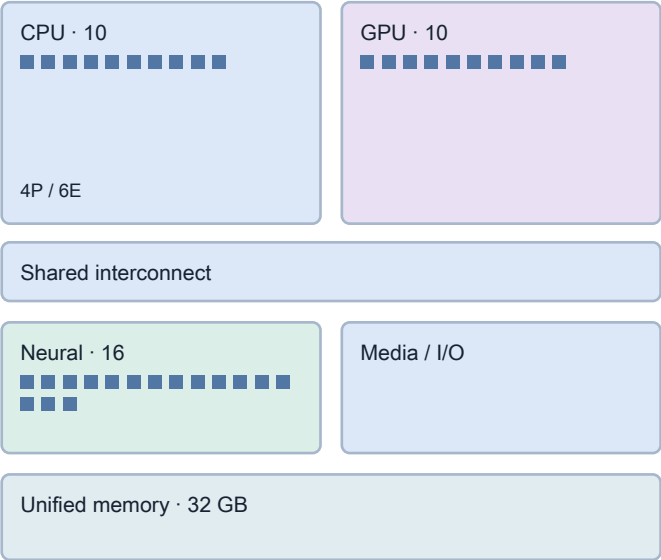
Functional schematic; not to physical scale.

# Architecture plates / 4

S: super · P: performance · E: efficiency. Every square = one core. GPU neural units (M5/M6) are not counted as additional GPU cores.

## M4

2024-05 / maximum configuration

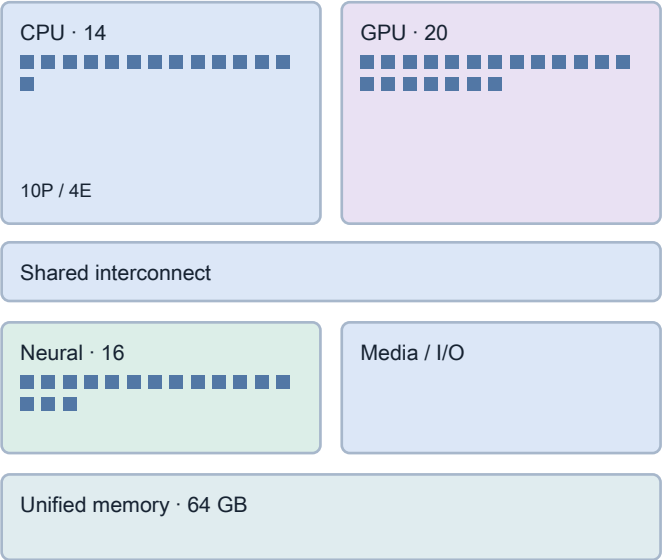


120 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M4 Pro

2024-10 / maximum configuration

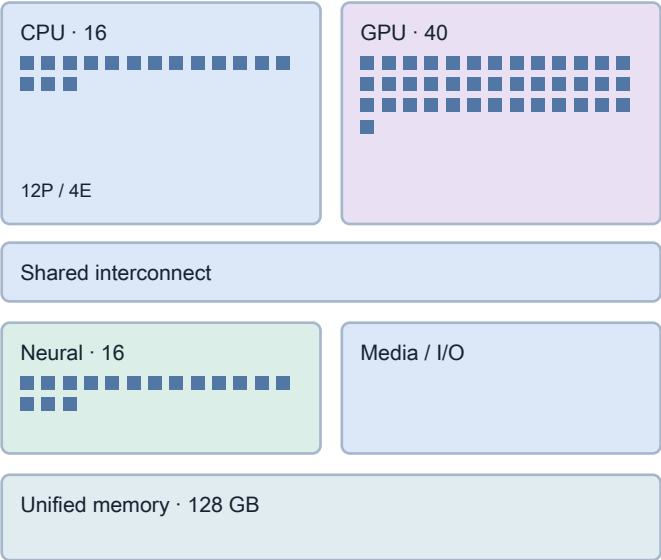


273 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M4 Max

2024-10 / maximum configuration

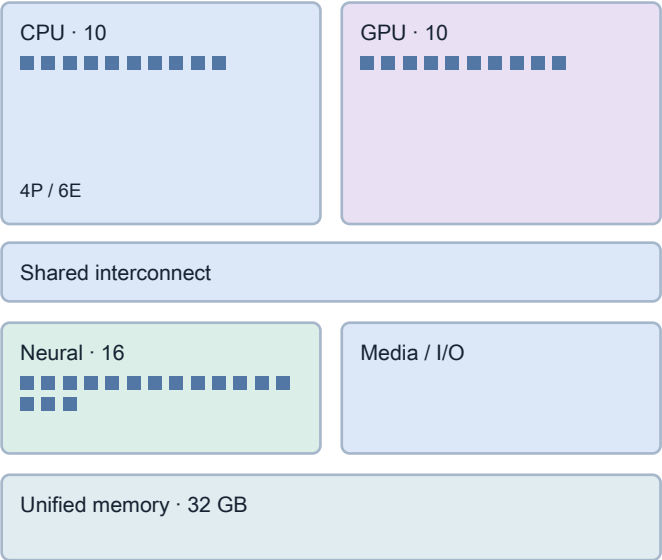


546 GB/s · 1 die(s)

Functional schematic; not to physical scale.

## M5

2025-10 / maximum configuration



153 GB/s · 1 die(s)

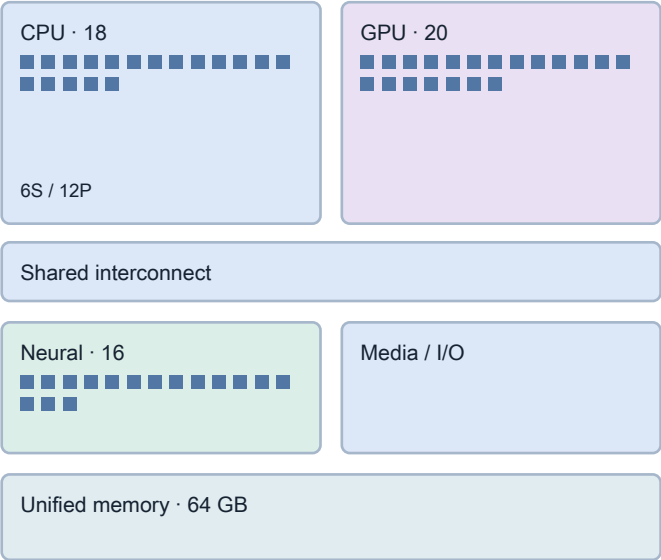
Functional schematic; not to physical scale.

# Architecture plates / 5

S: super · P: performance · E: efficiency. Every square = one core. GPU neural units (M5/M6) are not counted as additional GPU cores.

## M5 Pro

2026-03 / maximum configuration

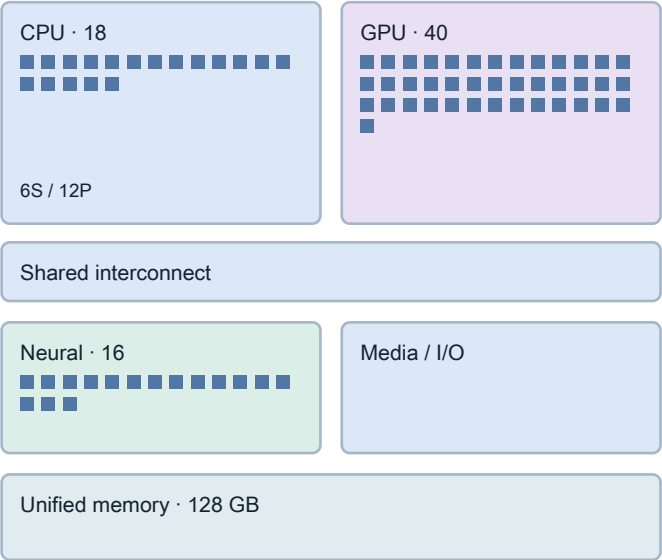


307 GB/s · 2 die(s)

Functional schematic; not to physical scale.

## M5 Max

2026-03 / maximum configuration

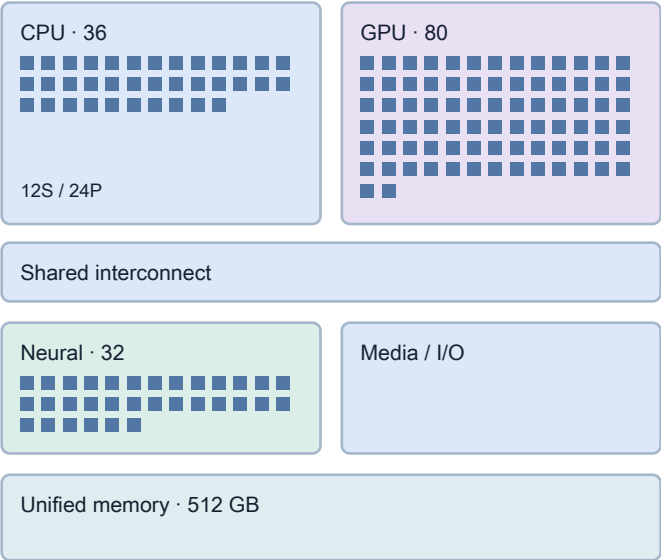


614 GB/s · 2 die(s)

Functional schematic; not to physical scale.

## M5 Ultra

2026-08 / maximum configuration

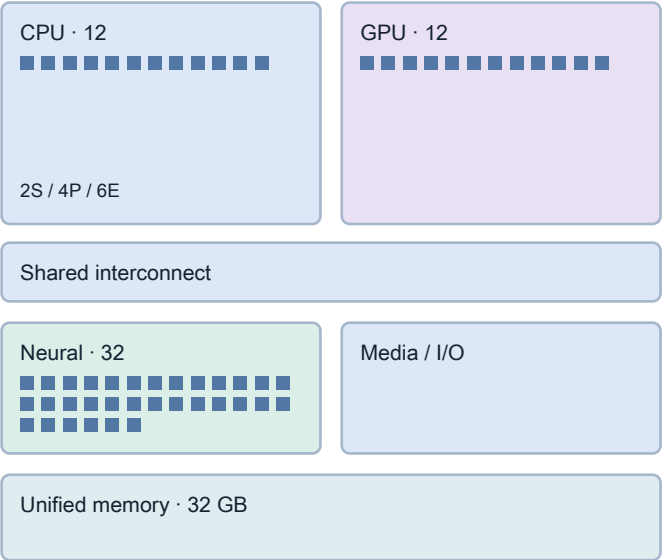


1200 GB/s · 4 die(s)

Functional schematic; not to physical scale.

## M6

2026-08 / maximum configuration

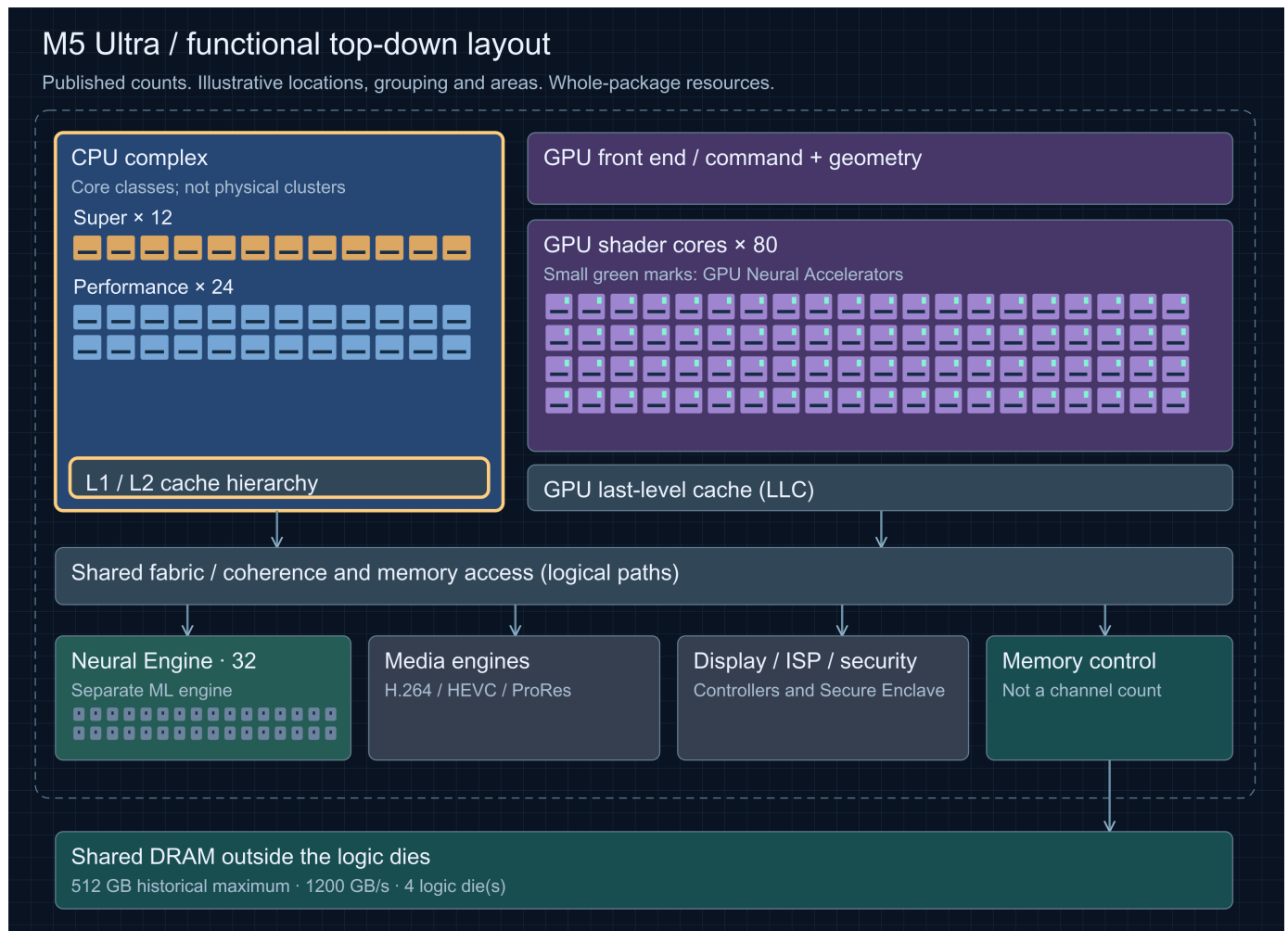


170 GB/s · 1 die(s)

Functional schematic; not to physical scale.

# Full-chip functional topology

Detailed functional schematic. The view explains structure, not measured physical geometry.



## Package versus die

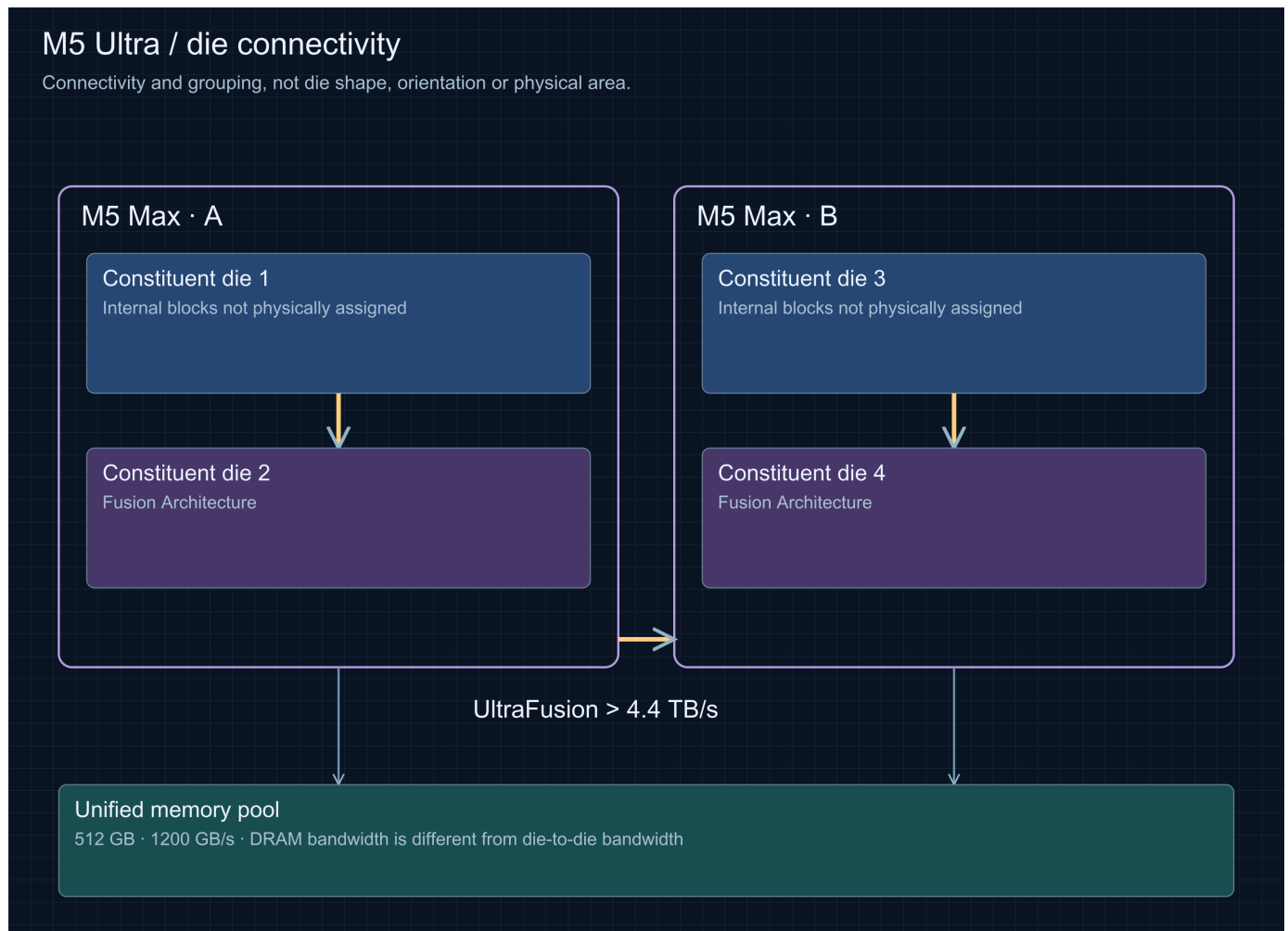
A package contains logic silicon, connections and memory components. A die is one fabricated piece of silicon. The map keeps unified memory outside the logic-die boundary. Physical locations, orientations and memory-package counts are not reconstructed from marketing pictures.

## Shared fabric and controllers

Blocks reach memory through shared system connections and memory controllers. The fabric strip is a logical connection, not a measured bus, mesh or crossbar. Aggregate DRAM bandwidth is not dedicated bandwidth for each block, and simultaneous clients can compete.

# Four-die connectivity: M5 Ultra

Detailed functional schematic. The view explains structure, not measured physical geometry.



## Fusion and UltraFusion

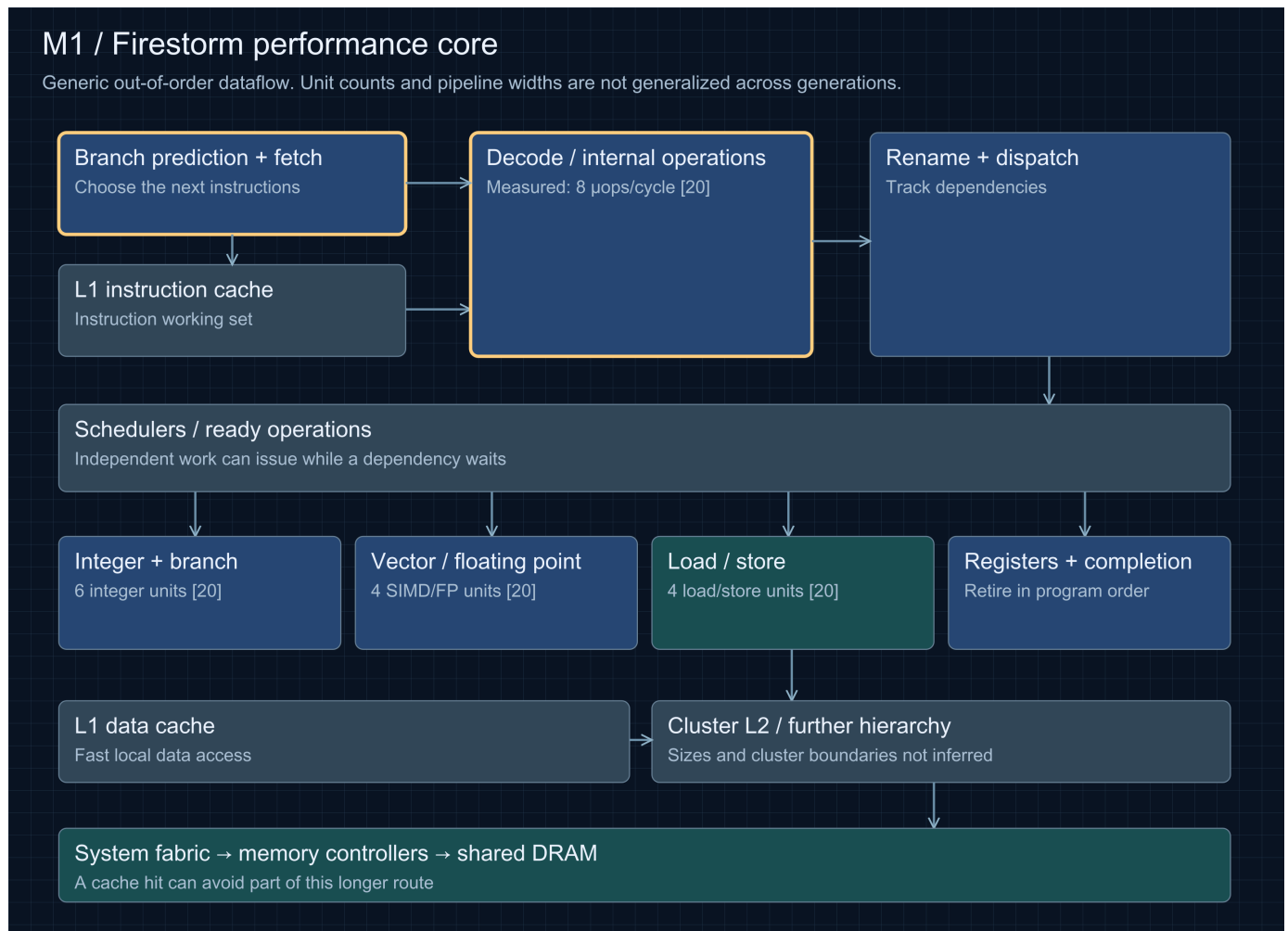
M1/M2/M3 Ultra combine two Max dies. M5 Pro/Max combine two dies with Fusion Architecture; M5 Ultra joins two dual-die Max chips, for four dies. This connectivity diagram asserts the grouping, not the physical orientation or a fully disclosed internal network.

## Unified DRAM

CPU, GPU and other engines can access a shared pool. The advantage is avoiding some transfers between separate CPU and GPU memory allocations. Cache ownership, synchronization, software buffers and device allocation limits still matter. The SSD is a separate storage device.

# Inside a CPU: M1 Firestorm

Detailed functional schematic. The view explains structure, not measured physical geometry.



## CPU front end

Instruction fetch and branch prediction keep the core supplied; decoding turns instructions into internal work. These are conceptual pipeline stages, not a claim that every generation has the same decoder width. For M1 only, independent Firestorm measurements report a pipeline limit of eight  $\mu$ ops per cycle.

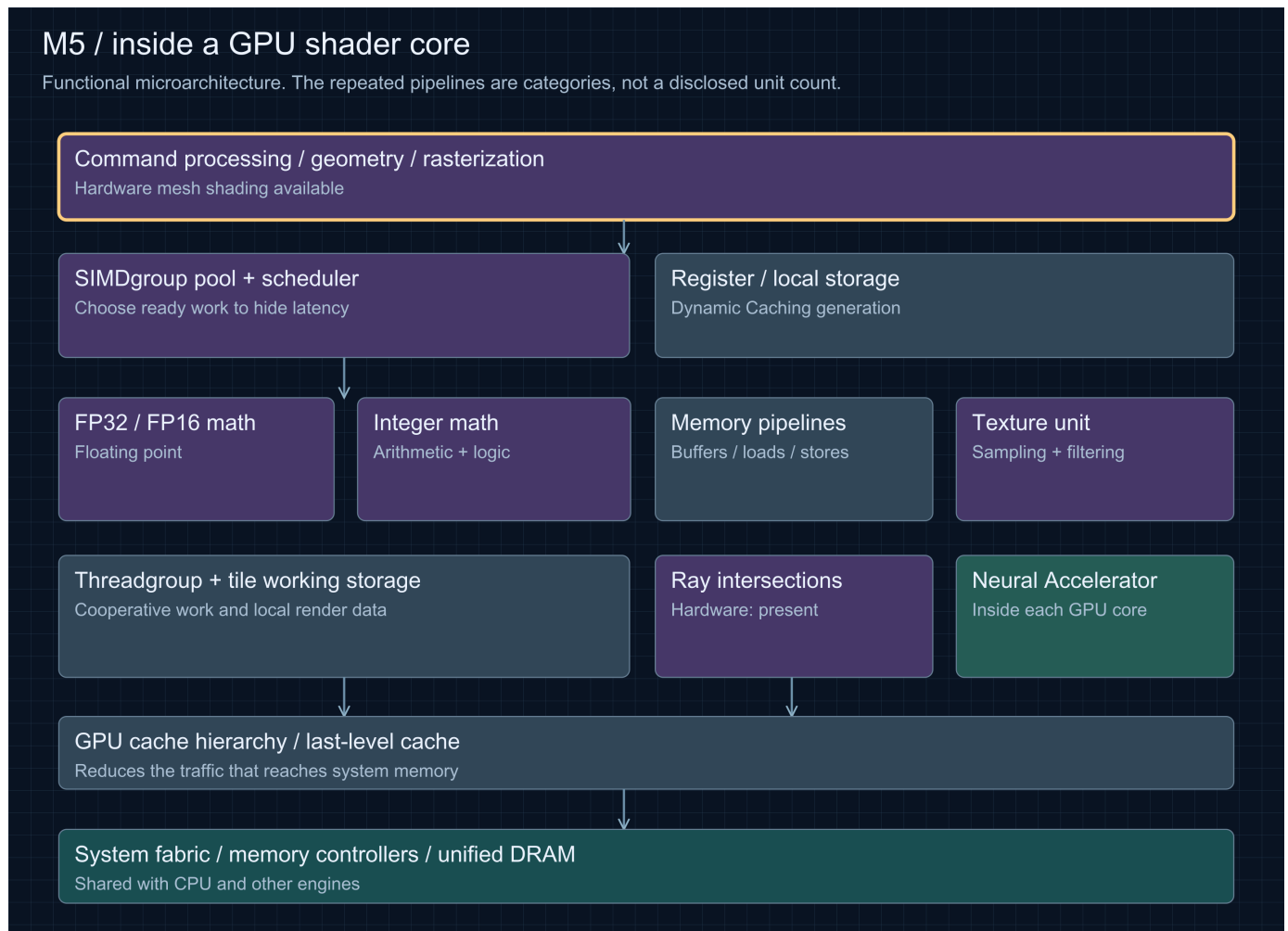
## Out-of-order execution

Renaming, scheduling and execution allow independent instructions to proceed while another waits. Integer arithmetic, branches, vector/floating-point work and loads/stores use different resources. The diagram names functions without inventing execution-unit counts for later Apple cores.

The numeric unit counts apply only to the M1 Firestorm performance core, from independent black-box measurements [20]. They do not describe the efficiency core or later generations.

# Inside a GPU: M5

Detailed functional schematic. The view explains structure, not measured physical geometry.



## Shader scheduling and execution

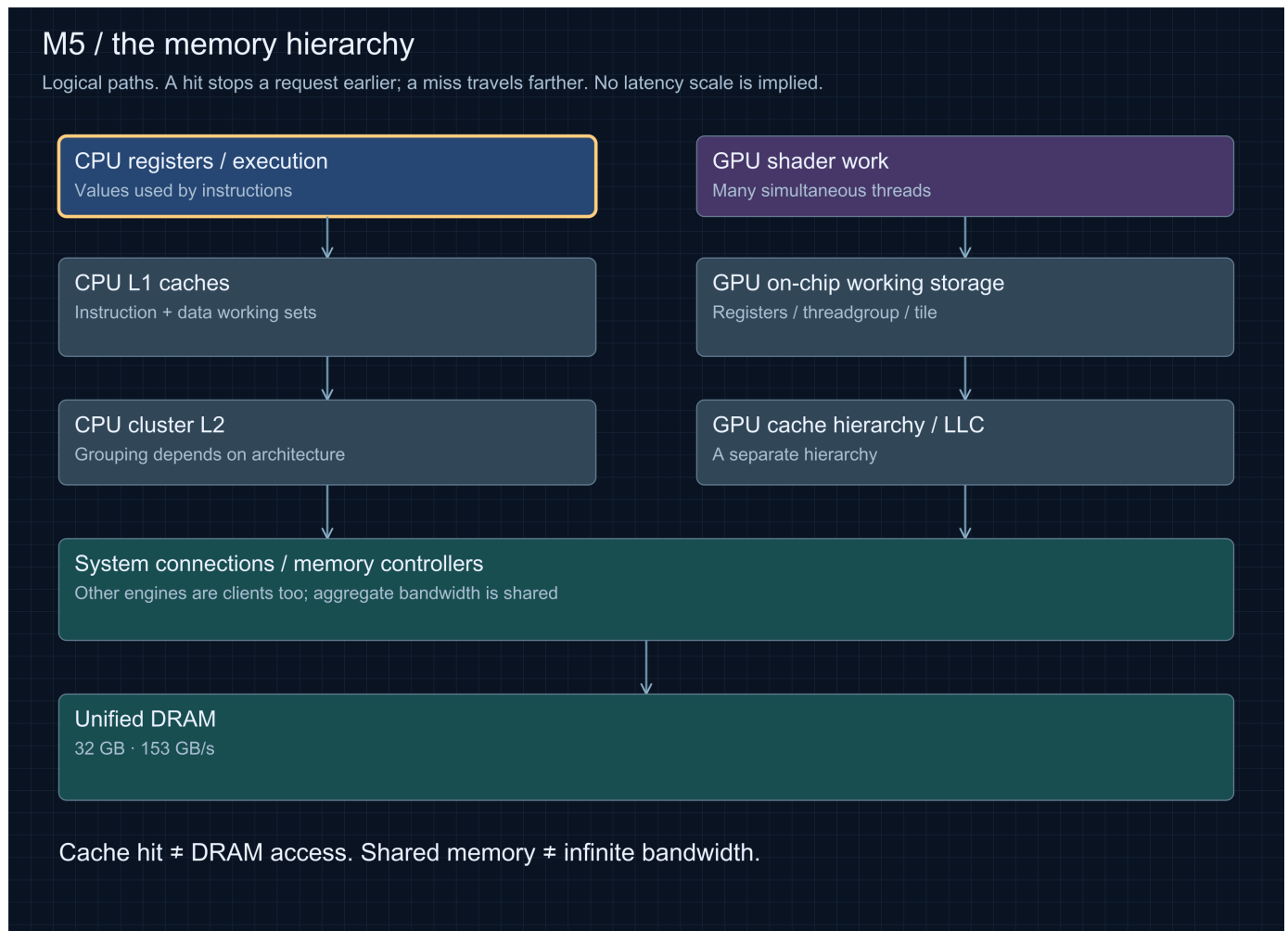
A shader core schedules SIMDgroups onto arithmetic and memory pipelines. Switching to ready work helps hide memory latency. Apple documents floating-point, integer and memory execution paths; their exact count and layout are not inferred here.

## GPU Neural Accelerator

M5 and M6 place a Neural Accelerator in each GPU core. These accelerate suitable AI computations through supported software. They belong to the GPU path and are not the independent Neural Engine. We show presence, not an undisclosed matrix-array size.

# Follow a memory request

Detailed functional schematic. The view explains structure, not measured physical geometry.



## CPU cache hierarchy

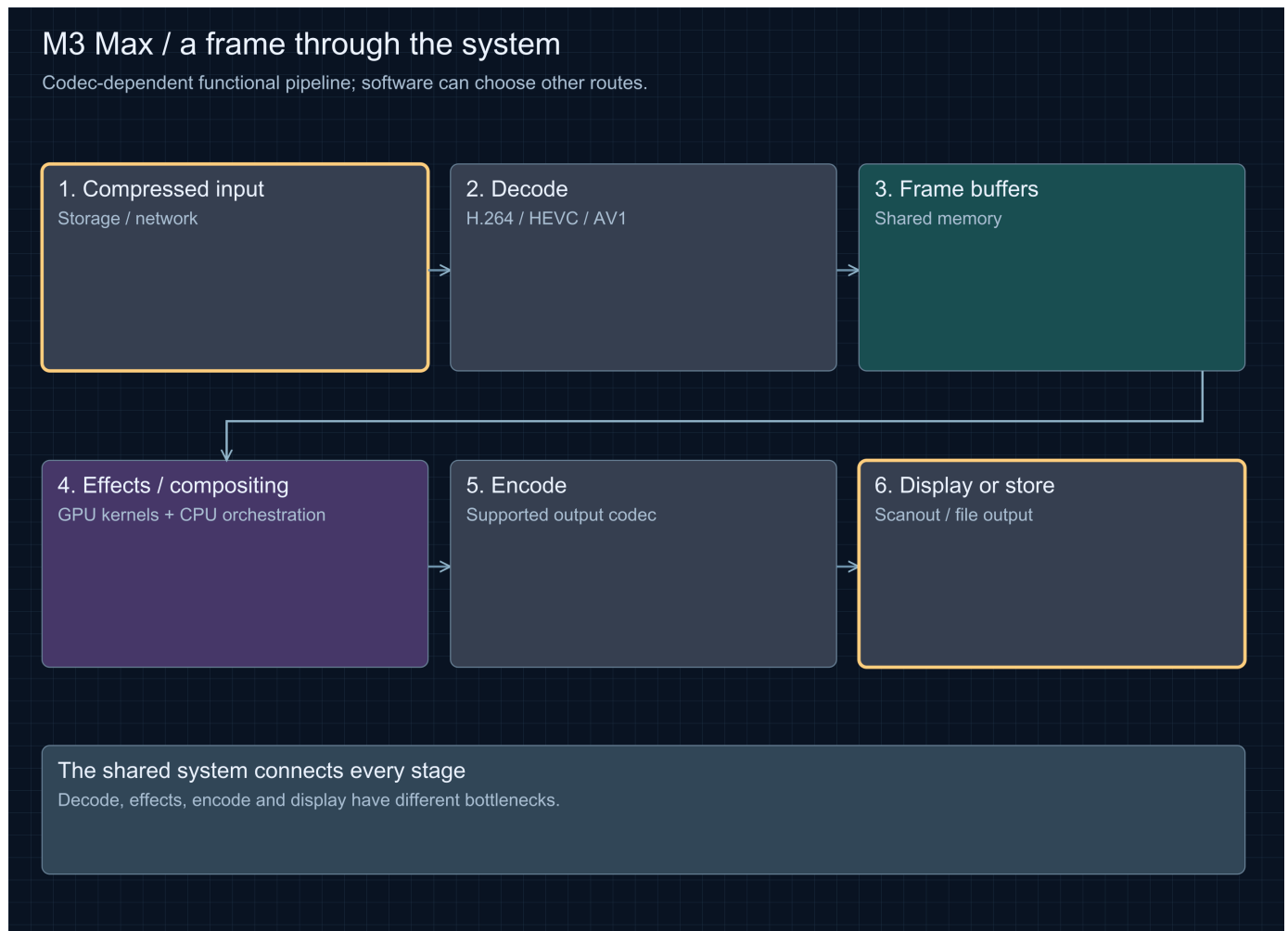
L1 instruction and data caches are near each core. A cluster-level L2 reduces traffic farther from the CPU. Grouping cores by type here is pedagogical: it does not assert an undocumented cluster count, cache size, associativity or cache-coherence protocol for M2–M6.

## GPU last-level cache

The GPU cache hierarchy reduces traffic to system memory. Apple's GPU counters distinguish accesses and misses; a cache miss may require a longer path. GPU LLC and CPU L2 are different structures. Neither should be confused with unified-memory capacity.

# Follow a video frame

Detailed functional schematic. The view explains structure, not measured physical geometry.



## Video dataflow

Compressed input is decoded into frames, effects transform them, and an encoder can produce compressed output. Codec support determines whether dedicated media blocks are used. GPU effects and CPU orchestration remain separate stages, and AV1 decoding is not AV1 encoding.

# Inside the topology / 1

## Package versus die

A package contains logic silicon, connections and memory components. A die is one fabricated piece of silicon. The map keeps unified memory outside the logic-die boundary. Physical locations, orientations and memory-package counts are not reconstructed from marketing pictures. [1] [3] [12] [13]

## CPU front end

Instruction fetch and branch prediction keep the core supplied; decoding turns instructions into internal work. These are conceptual pipeline stages, not a claim that every generation has the same decoder width. For M1 only, independent Firestorm measurements report a pipeline limit of eight  $\mu$ ops per cycle. [20]

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Renaming, scheduling and execution allow independent instructions to proceed while another waits. Integer arithmetic, branches, vector/floating-point work and loads/stores use different resources. The diagram names functions without inventing execution-unit counts for later Apple cores. [20]

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L1 instruction and data caches are near each core. A cluster-level L2 reduces traffic farther from the CPU. Grouping cores by type here is pedagogical: it does not assert an undocumented cluster count, cache size, associativity or cache-coherence protocol for M2–M6. [1] [20]

## GPU command and geometry front end

Metal command buffers describe work. Compute and vertex command processors feed the GPU; geometry and rasterization prepare fragment work. Hardware mesh shading is a capability boundary introduced with M3, but supported software must request the appropriate pipeline. [7] [18]

## Shader scheduling and execution

A shader core schedules SIMDgroups onto arithmetic and memory pipelines. Switching to ready work helps hide memory latency. Apple documents floating-point, integer and memory execution paths; their exact count and layout are not inferred here. [18]

# Inside the topology / 2

## Registers, tile and threadgroup storage

Registers hold per-thread values; threadgroup storage supports cooperation; tile storage holds local rendering data. M3's Dynamic Caching changes allocation and reuse of on-chip storage. It does not increase installed RAM or make memory latency disappear. [7] [18]

## Texture sampling

Texture hardware serves filtering and texture access. This is distinct from arithmetic throughput. A shader can be limited by sampling, memory traffic, occupancy or math, so adding shader cores alone need not solve the bottleneck. [18] [19]

## Ray-intersection acceleration

From M3 onward, hardware accelerates ray-intersection work. Shaders still construct and consume the surrounding rendering algorithm. The presence of a ray-tracing unit is not a promise that every scene or game becomes faster by one fixed ratio. [7] [18]

## GPU Neural Accelerator

M5 and M6 place a Neural Accelerator in each GPU core. These accelerate suitable AI computations through supported software. They belong to the GPU path and are not the independent Neural Engine. We show presence, not an undisclosed matrix-array size. [11] [12] [13]

## GPU last-level cache

The GPU cache hierarchy reduces traffic to system memory. Apple's GPU counters distinguish accesses and misses; a cache miss may require a longer path. GPU LLC and CPU L2 are different structures. Neither should be confused with unified-memory capacity. [19]

## Shared fabric and controllers

Blocks reach memory through shared system connections and memory controllers. The fabric strip is a logical connection, not a measured bus, mesh or crossbar. Aggregate DRAM bandwidth is not dedicated bandwidth for each block, and simultaneous clients can compete. [1] [19]

# Inside the topology / 3

## Unified DRAM

CPU, GPU and other engines can access a shared pool. The advantage is avoiding some transfers between separate CPU and GPU memory allocations. Cache ownership, synchronization, software buffers and device allocation limits still matter. The SSD is a separate storage device. [1] [19]

## The separate Neural Engine

This is a dedicated ML engine, not a set of GPU shader cores. M1–M5 non-Ultra chips use 16 cores in this catalogue; Ultra chips use 32. M6 has two 16-core engines. Dispatch depends on the model, operations and framework, not simply on the presence of neural hardware. [1] [13] [17]

## Video dataflow

Compressed input is decoded into frames, effects transform them, and an encoder can produce compressed output. Codec support determines whether dedicated media blocks are used. GPU effects and CPU orchestration remain separate stages, and AV1 decoding is not AV1 encoding. [2] [7] [15]

## Display, storage, security and I/O

Display scanout, image processing, storage control, external I/O and Secure Enclave are specialized functions. This diagram groups them to explain the system. Counts, bandwidth and port/display support must be checked for the actual device. [1] [15] [21] [22]

## Fusion and UltraFusion

M1/M2/M3 Ultra combine two Max dies. M5 Pro/Max combine two dies with Fusion Architecture; M5 Ultra joins two dual-die Max chips, for four dies. This connectivity diagram asserts the grouping, not the physical orientation or a fully disclosed internal network. [3] [6] [8] [12] [13]

# The building blocks

Select these blocks in the interactive edition to trace workloads.

## CPU

Runs general code: control flow, branches, operating-system work, and many latency-sensitive tasks. More cores help independent work; a serial dependency still waits. Performance, efficiency and super cores are not interchangeable units across generations.

## GPU

Runs many similar operations in parallel for graphics and compute. Core count is not a cross-generation benchmark. Memory traffic, shader design, ray-tracing support and software determine which work benefits.

## Neural Engine

A dedicated machine-learning engine. Core ML can use CPU, GPU and Neural Engine according to the model and supported operations. Sixteen Neural Engine cores do not equal sixteen GPU cores, and peak TOPS do not measure end-to-end application latency.

## Unified memory

A shared pool accessible to the engines. Capacity is how much fits; bandwidth is how quickly data can move. Sharing can avoid copies between separate CPU and GPU pools, but does not remove cache misses, synchronization, allocation limits or bandwidth contention.

## Media engines

Fixed-function blocks accelerate supported video codecs. Decode, encode, effects and display are different stages. A fast ProRes path does not imply the same advantage for an unsupported codec or a GPU-heavy effect. Base M1 lacks the later dedicated ProRes acceleration.

## Interconnect & packaging

The fabric moves data among blocks; UltraFusion connects dies in Ultra chips. Inter-die bandwidth is not DRAM bandwidth. M1–M3 Ultra use two Max dies. M5 Pro/Max use two-die Fusion; M5 Ultra combines two of those Max chips into four dies.

## Security & I/O

Secure Enclave, storage, display and peripheral controllers are part of the system story. Actual ports, external-display limits and sustained power depend on the device, not only the chip name. Unified memory is not the SSD.

# Generation notes / 1

## M1 · 2020–2022

### The system becomes the chip

M1 brought the CPU, GPU, Neural Engine and controllers into a tightly integrated Mac system. Its four performance and four efficiency CPU cores share one memory pool with other engines. Integration is about reducing data movement as well as doing arithmetic.

Pro and Max expanded the resources in 2021; Ultra joined two Max dies in 2022. M1 Pro and M1 Max have the same maximum CPU core count, but Max doubles the maximum GPU count and memory bandwidth. This is a useful example of a tier upgrade that does not simply double everything.

Try M1 Pro against M1 Max: CPU counts stay at 10, while GPU counts rise from 16 to 32.

[\[1\] M1](#) [\[2\] M1 Pro / Max](#) [\[3\] M1 Ultra](#)

## M2 · 2022–2023

### More room, familiar structure

M2 refined the 5 nm generation. The base chip kept eight CPU cores while increasing the maximum GPU count to ten, unified memory to 24 GB, and bandwidth to 100 GB/s. ProRes hardware also reached the base tier.

M2 Pro and Max expanded to twelve CPU cores, while M2 Ultra paired two Max dies. The 192 GB advertised Ultra ceiling increased the scale of in-memory work. A larger memory pool can make a previously impossible workload possible without guaranteeing lower latency for small tasks.

Compare M2 Max with M2 Ultra: doubling resources helps only if the application can use them.

[\[4\] M2](#) [\[5\] M2 Pro / Max](#) [\[6\] M2 Ultra](#)

# Generation notes / 2

## M3 · 2023–2025

### A new graphics capability boundary

The first 3 nm M-series generation introduced Dynamic Caching, hardware-accelerated ray tracing and mesh shading. These are capabilities of the graphics architecture, not just increases in core count. AV1 hardware decoding joined the media features.

The family is not a monotonic ladder on every specification: M3 Pro has 150 GB/s of bandwidth against M2 Pro's 200 GB/s. M3 Ultra arrived in 2025, after M4, with up to 32 CPU and 80 GPU cores. Its original advertised memory ceiling was 512 GB; later configuration availability can differ.

Try M2 Pro against M3 Pro: the newer design adds graphics features while the published bandwidth decreases.

[\[7\] M3 / Pro / Max](#) [\[8\] M3 Ultra](#) [\[14\] MacBook Pro • 2023](#) [\[15\] Mac Studio • 2025](#)

## M4 · 2024

### Faster engines, wider memory paths

M4 debuted in iPad Pro before arriving in Macs. Its second-generation 3 nm process accompanied CPU, GPU and Neural Engine improvements. The Mac family reaches 32 GB on base M4, 64 GB on Pro and 128 GB on Max. These are family ceilings, not the memory fitted to every device.

M4 Pro and Max introduced Thunderbolt 5 support on Mac and increased memory bandwidth to 273 and up to 546 GB/s. The 32-core-GPU M4 Max instead has 410 GB/s. There is no M4 Ultra in this verified catalogue; generation names do not promise a complete set of tiers.

Compare M4 with M3 Ultra: a newer base chip and an older desktop-scale chip answer different needs.

[\[9\] M4](#) [\[10\] M4 / Pro / Max](#) [\[15\] Mac Studio • 2025](#)

# Generation notes / 3

## M5 · 2025–2026

### AI acceleration moves inside GPU cores

M5 adds a Neural Accelerator to each GPU core. These units are separate from the Neural Engine. A workload must reach the appropriate hardware through supported software; the presence of an accelerator is not a universal multiplier for every AI application. Base M5 raises memory bandwidth to 153 GB/s.

Pro and Max use a two-die Fusion Architecture, with up to six super cores and twelve performance cores. M5 Ultra joins two dual-die Max chips: four dies, up to 36 CPU cores and 80 GPU cores. Apple specifies up to 512 GB of memory and 1.2 TB/s bandwidth. Packaging topology and the programming model are different questions.

Compare M5 Max and M5 Ultra, then switch the diagram from GPU to memory to see what scales.

[\[11\] M5](#) [\[12\] M5 Pro / Max](#) [\[13\] M6 / M5 Ultra](#)

## M6 · 2026

### A three-class CPU and two neural engines

Announced on 25 August 2026, M6 introduces Apple's 2 nm process generation. The twelve-core CPU combines two super, four performance and six efficiency cores. The GPU has twelve cores with Neural Accelerators; Apple describes a Dual 16-core Neural Engine.

The base chip supports up to 32 GB and 170 GB/s. Those figures do not make it a replacement for an Ultra when the workload needs hundreds of gigabytes. Only the announced base M6 is included; this atlas does not invent Pro, Max or Ultra specifications.

Select M6 against M5 Ultra: generation and scale point in different directions.

[\[13\] M6 / M5 Ultra](#)

# Capacity is a gate; bandwidth is a rate

A teaching model, not a measured hardware benchmark.

A dense model with  $P$  billion parameters stored at  $q$  bits per weight needs approximately  $P \times q / 8$  decimal GB for the weights alone. Add runtime data, quantization metadata, activations and the KV cache, and leave space for the operating system. The experiment uses an explicit, adjustable reserve rather than pretending that all installed RAM is free.

For a deliberately simplified, batch-one decode model, assume each generated token reads the active weights once. The bandwidth-only ceiling is bandwidth divided by weight size. This excludes compute, attention, cache traffic and framework overhead. It is a teaching upper bound, never a measured tokens-per-second result. MoE models require active-weight accounting, and batching changes reuse.

Example: 8 billion weights at 4 bits occupy 4 GB before overhead. At 100 GB/s, the idealized weight-streaming ceiling is 25 tokens/s. It is not a prediction for a real Mac.

## Weight memory at four bits per parameter

8B → 4 GB weights alone

32B → 16 GB weights alone

70B → 35 GB weights alone

120B → 60 GB weights alone

200B → 100 GB weights alone

Try 70B at 4 bits with an 8 GB reserve: 43 GB is required by this simplified capacity budget. A 32 GB ceiling fails; 64 GB passes the capacity test, without promising that the software can allocate or run it well.

# Why doubling cores rarely halves time

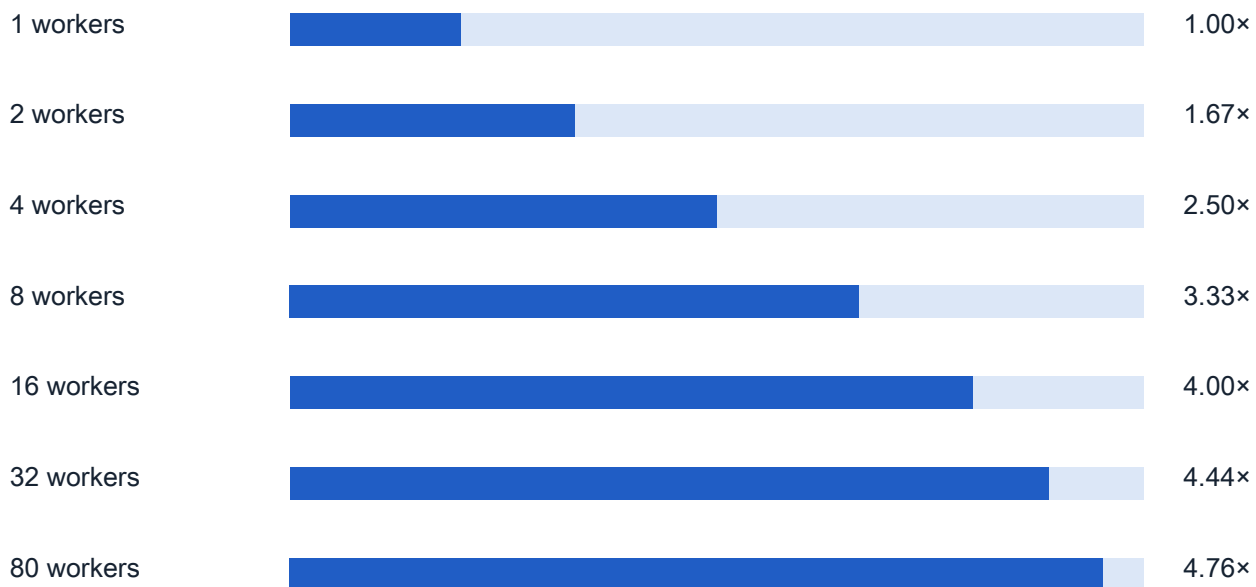
A teaching model, not a measured hardware benchmark.

Amdahl's law separates a serial fraction  $s$  from parallel work. With  $N$  identical ideal workers, speedup is  $1 / (s + (1 - s) / N)$ . When 20% is serial, even infinitely many workers cannot exceed 5×. Real scheduling, unequal core types and memory contention reduce the benefit further.

This experiment uses hypothetical equal workers rather than assigning invented performance scores to Apple cores. It explains scaling; it cannot rank M1 against M6. Single-thread performance, acceleration, thermal headroom and algorithm changes are separate effects.

Example: at 20% serial work, 8 workers give 3.33× and 16 give 4×. Twice the workers produce only a 20% speedup.

## At 20% serial work



# Read benchmarks as experiments

Ask which device, memory configuration, power mode, application, precision, model and workload were tested. A short burst and a sustained render measure different conditions. Fanless and actively cooled systems with the same chip can diverge over time.

An Apple “up to” claim is a result under its stated test conditions, not a universal ratio. Do not chain ratios from different tests, equate TOPS at different precisions, or convert core count into a synthetic overall performance score. This atlas therefore charts specifications, not an invented benchmark ranking.

Upgrade around the bottleneck: capacity for work that cannot fit; bandwidth for streaming; GPU features for supported graphics; CPU latency for serial code; media engines for supported codecs. Check the application and the complete device before choosing.

## GPU rendering

Check renderer support, scene memory and feature requirements before comparing GPU counts.

## Local AI

Distinguish prompt processing from token generation, and dense models from mixture-of-experts models.

## Video and audio

Codec acceleration and real-time latency constraints matter as much as peak throughput.

# Sources and scope / 1

Primary sources checked 19 September 2026. Links are clickable.

## [1] M1

<https://www.apple.com/newsroom/2020/11/apple-unleashes-m1/>

## [2] M1 Pro / Max

<https://www.apple.com/newsroom/2021/10/introducing-m1-pro-and-m1-max-the-most-powerful-chips-apple-has-ever-built/>

## [3] M1 Ultra

<https://www.apple.com/newsroom/2022/03/apple-unveils-m1-ultra-the-worlds-most-powerful-chip-for-a-personal-computer/>

## [4] M2

<https://www.apple.com/newsroom/2022/06/apple-unveils-m2-with-breakthrough-performance-and-capabilities/>

## [5] M2 Pro / Max

<https://www.apple.com/newsroom/2023/01/apple-unveils-m2-pro-and-m2-max-next-generation-chips-for-next-level-workflows/>

## [6] M2 Ultra

<https://www.apple.com/newsroom/2023/06/apple-introduces-m2-ultra/>

## [7] M3 / Pro / Max

<https://www.apple.com/newsroom/2023/10/apple-unveils-m3-m3-pro-and-m3-max-the-most-advanced-chips-for-a-personal-computer/>

## [8] M3 Ultra

<https://www.apple.com/newsroom/2025/03/apple-reveals-m3-ultra-taking-apple-silicon-to-a-new-extreme/>

# Sources and scope / 2

Primary sources checked 19 September 2026. Links are clickable.

## [9] M4

<https://www.apple.com/newsroom/2024/05/apple-introduces-m4-chip/>

## [10] M4 / Pro / Max

<https://www.apple.com/newsroom/2024/10/apple-introduces-m4-pro-and-m4-max/>

## [11] M5

<https://www.apple.com/newsroom/2025/10/apple-unleashes-m5-the-next-big-leap-in-ai-performance-for-apple-silicon/>

## [12] M5 Pro / Max

<https://www.apple.com/newsroom/2026/03/apple-debuts-m5-pro-and-m5-max-to-supercharge-the-most-demanding-pro-workflows/>

## [13] M6 / M5 Ultra

<https://www.apple.com/newsroom/2026/08/apple-introduces-m6-and-m5-ultra-for-a-big-leap-in-performance-and-ai-compute/>

## [14] MacBook Pro • 2023

<https://support.apple.com/en-au/117737>

## [15] Mac Studio • 2025

<https://support.apple.com/en-au/122211>

## [16] Metal GPU families

<https://developer.apple.com/metal/Metal-Feature-Set-Tables.pdf>

# Sources and scope / 3

Primary sources checked 19 September 2026. Links are clickable.

## [17] Core ML

<https://developer.apple.com/documentation/coreml>

## [18] GPU architecture • Apple Tech Talk

<https://developer.apple.com/videos/play/tech-talks/111375/>

## [19] GPU caches and counters • WWDC20

<https://developer.apple.com/videos/play/wwdc2020/10603/>

## [20] Firestorm microarchitecture • measured research

<https://dougallj.github.io/applecpu/firestorm.html>

## [21] Mac mini specifications • M6

<https://www.apple.com/mac-mini/specs/>

## [22] Mac Studio specifications • M5 Ultra

<https://www.apple.com/mac-studio/specs/>

## Editorial method

Specifications use Apple launch releases and technical pages; the M3 and M4 Max bandwidth exceptions are checked against support specifications. Published maximum memory describes capability over time, not stock availability. Models and diagrams are original explanatory constructions. No fabricated benchmarks, die areas, transistor layouts or unreleased tier specifications are included.

The Japanese edition is AI-translated and checked for technical consistency against the English dataset. The M-series motion coprocessors are outside scope.